LOKESH RAM HARIRAM

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SKILLS

AI/ML & LLM Engineering: Python, LLMs, Generative AI, Amazon Bedrock, Azure OpenAI, RAG, AI Agents, Agentic AI, LangChain, Embeddings, Vector Search, Prompt Engineering, Tool/Function Calling, LLM Evaluation, Guardrails

Machine Learning & Predictive Analytics: Scikit-learn, XGBoost, ARIMA, Predictive Modeling, Time-Series Forecasting, Feature Engineering, Model Evaluation, Model Monitoring

Full-Stack Development: JavaScript, TypeScript, React.js, Node.js, Express.js, REST APIs, Serverless APIs, Authentication & Authorization, Claude Code, GitHub Copilot

Data Engineering & Databases: SQL, T-SQL, PostgreSQL, Data Modeling, ETL/ELT Pipelines, Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Microsoft Fabric

Cloud, DevOps & Testing: Microsoft Azure, AWS, Docker, Kubernetes, Git, GitHub, CI/CD, Automated Testing, Unit Testing, Integration Testing, Azure DevOps, Jira, Confluence, Notion

PROFESSIONAL EXPERIENCE

Applied AI/ML Engineer - *American College of Emergency Physicians (ACEP), Irving, TX* Aug 2025 – Jan 2026

- Built AskEMDI**,** an LLM-powered natural-language-to-SQL platform using Amazon Bedrock and Anthropic Claude, translating emergency physicians and researchers questions into database queries against structured emergency-medicine data.
- Partnered with clinical and operations stakeholders to translate research questions into AI workflows, gather requirements, and refine query logic based on stakeholder feedback.
- Engineered ETL/ELT pipelines using Azure Data Factory, Azure Databricks, and Azure Synapse Analytics, alongside Python/SQL data-quality checks to identify missing values, inconsistent records, and clinical coding issues across a 23M+ encounter dataset of de-identified clinical data.

Research Assistant (Machine Learning) - *SEAR LAB, Arlington, TX* Aug 2024 – July 2025

- Developed XGBoost and ARIMA models on time-series sensor data for forecasting and evaluated predictive accuracy using RMSE/MAE on a chronological train/test split.
- Built Python and SQL pipelines to clean, transform, and engineer features from 500K+ sensor records, producing model-ready datasets for forecasting and operational analysis.
- Automated reporting with Python, Matplotlib, and Power BI, reducing KPI reporting time by 40%; refined reporting metrics with research and engineering stakeholders based on their feedback.

EDUCATION

The University of Texas at Arlington

Master of Science in Information Systems

Sathyabama University

Bachelor of Science in Computer Science

PROJECT EXPERIENCE

E-Commerce Pricing Risk Detection System

Tech Stack: Python, XGBoost, Sentence Transformers, PostgreSQL, FastAPI, Streamlit, Docker

- Built a pricing-risk detection pipeline using Amazon Reviews 2023 Electronics metadata, engineering category-relative price deviations, discount features, product attributes, and Sentence Transformer embeddings to generate product-level risk scores with XGBoost.
- Integrated PostgreSQL to store product metadata and risk scores; developed a FastAPI scoring service and Streamlit dashboard to prioritize suspicious listings for human review.
- Created a synthetic-anomaly benchmark by injecting pricing anomalies into real product records; on a held-out test set, achieved 80% recall at 75% precision while flagging 15% of listings for human review.

API Integration s Contract Verification Agent

Tech Stack: Python, Claude API, Shippo API, FastAPI, Docker, JSON Schema

- Built a Claude-powered API integration agent to analyze Shippo documentation for **10 endpoints**, generating endpoint clients, request mappings, response schemas, and executable integration tests.
- Implemented a bounded generate–execute–diagnose–repair workflow that validates requests against the Shippo test environment, JSON Schemas, retrying for predefined failures such as missing fields, invalid parameter mappings.
- Generated endpoint-level verification reports covering test outcomes, schema mismatches, repair attempts, and unresolved failures, enabling engineers to identify integration issues and review endpoint readiness before handoff.